ENGR 183EW

DIS 1E

An Electrified Economy

Can the national goal of electrifying the economy be achieved?

—

By Pranav Varmaraja, Sahil Gosain, Supavit (Troy) Pitaknarongphorn, Yash Kothari

Table of Content

Table of Content **2**
0. Executive Summary **4**
0.1 Overview of the Problem 4
0.2 Significance of Achieving an Electrified Economy 4
0.3 Technological and Ethical Challenges 5
0.4 Proposed Approach 6
0.5 High-Level Conclusion 8
1. Introduction **8**
1.1 Methodology 8
1.2 Work Breakdown Structure (WBS) 9
1.3 Responsibility Assignment Matrix (RAM) Chart 9
2. Problem Statement and Background **10**
2.1 Background on Electrifying the Economy 10
2.2 National Goals for Electrification and Decarbonization 11
2.3 Current Energy Infrastructure and Limitations 13
2.3.1 Dependence on Fossil Fuels 13
2.3.2 Aging Grid Infrastructure 13
2.3.3 Centralized Power Generation Model 14
2.3.4 Vulnerability to Extreme Weather 14
2.3.5 Lack of Energy Source Autonomy 14
2.3.6 Insufficient Energy Storage 15
3. Technological Issues **16**
3.1 Renewable Energy 16
3.2 Grid Infrastructure 17
3.3 Battery Technology 18
3.4 Electric Vehicle 19
3.5 Material Supply Chain and Mining 20
3.6 Costs and Scalability 21
4. Ethical and Societal Issues **22**
4.1 Differences in Transitioning Renewable Electrification in Rural America 22
4.2 Inequities in Urban Electrification 24
4.2.1 Hindrances in Electric Vehicle (EV) adoption 24

4.2.2 Difficulties in Electrifying Urban Buildings....25
4.3 Societal Health-Related Concerns of Electrification....26
4.4 Supply Chain Tolls....27
5. Recommended Solution....27
5.1 Technical Recommendations....27
5.1.1 Modernization of the Electric Grid....27
5.1.2 Investment in Renewable Energy Infrastructure....28
5.1.3 Development of Energy Storage Solutions....28
5.2 Ethical Recommendations....29
5.2.1 Promoting Equity in Energy Access....29
5.2.2 Ethical Governance and Policy Reform....31
5.2.3 Invest in Education....31
6. Conclusion....31
7. References....33

0. Executive Summary

0.1 Overview of the Problem

The national goal of electrifying the economy aims to transition the United States from a fossil fuel-dependent energy system to one primarily powered by electricity generated from clean, renewable sources such as solar, wind, and hydroelectric power. This ambitious objective includes specific targets set by the Department of Energy: achieving 100% clean electricity by 2035 and attaining net-zero economy-wide greenhouse gas (GHG) emissions by no later than 2050.

Achieving such a goal involves a comprehensive overhaul of various sectors traditionally reliant on fossil fuels. For example, transportation would see a shift from gasoline and diesel-powered vehicles to electric vehicles (EVs), necessitating a complete transformation of fuel infrastructure from gas stations to charging stations as well as technological improvements within EVs themselves. Furthermore, industrial processes would adapt to electrified machinery, and residential and commercial buildings would undergo significant renovations to eliminate dependence on natural gas and other non-renewable energy sources.

Recent events, however, have raised concerns about the feasibility and pace of this transition. California's near miss with an energy crisis last summer—prompting emergency text messages requesting electricity use reduction—highlighted the limitations of the current electric grid under increased demand. In the case of electrifying transportation, the performance challenges of EVs during severe winter blizzards, including reduced battery efficiency and range, have questioned the reliability of EV technology under extreme conditions. The insufficient number of EV charging stations, despite federal initiatives to expand infrastructure, further complicates the rapid rollout of electrified transportation. These issues underscore some of the fundamental issues that may arise when attempting to electrify the economy rapidly.

0.2 Significance of Achieving an Electrified Economy

Transitioning to an electrified economy is critical for multiple reasons. Environmentally, reducing reliance on fossil fuels will significantly decrease greenhouse gas emissions, mitigating the impacts of climate change and global warming. Improved air quality

resulting from lower emissions will lead to public health benefits by reducing health risks associated with pollution-related illnesses. Energy independence is another significant advantage; relying on domestic renewable energy sources enhances national security by reducing dependence on foreign oil and fossil fuel imports. Economically, the shift stimulates growth through innovation in clean energy technologies, creating new job opportunities in emerging industries. Furthermore, an electrified economy promotes sustainability by ensuring the long-term availability of resources without compromising the needs of future generations.

0.3 Technological and Ethical Challenges

Despite the clear benefits, several technological and ethical challenges hinder the rapid achievement of an electrified economy. One major challenge is the limitations of the electric grid. The current grid infrastructure is aging and was designed for centralized, fossil fuel-based power generation. Approximately 70% of transmission lines are over 25 years old, approaching the end of their useful life. The grid lacks the capacity and resilience to handle increased demand from widespread electrification and to effectively integrate distributed electricity obtained via renewable energy sources.

Another significant hurdle is the current lack of sufficient EV infrastructure. First and foremost, there is an insufficient number of EV charging stations, particularly in rural and underserved areas. Furthermore, the existing charging network is plagued by reliability issues, with studies showing that about 25% of public chargers are non-functional. This infrastructure gap poses a substantial barrier to the widespread adoption of EVs.

Technological reliability and performance issues also pose challenges. EVs often face performance problems under extreme weather conditions. Cold temperatures can reduce battery efficiency and vehicle range, raising concerns about the reliability of EV technology during winter blizzards and other severe weather events.

Cost and implementation challenges further complicate the transition. Upgrading the electric grid, expanding renewable energy production, and building out EV infrastructure require substantial financial investment and coordination across multiple sectors. High initial costs for consumers, such as purchasing EVs and retrofitting homes for electrification, can also be prohibitive to adoption in some cases.

Another potential issue that may come with the transition to a completely electrified economy is the exacerbation of existing societal inequities. Low-income communities

might be disproportionately affected due to higher energy costs, lack of access to EVs and charging infrastructure, and potential job losses in fossil fuel industries. Energy poverty remains a significant issue, with a significant number of families in the U.S. struggling to afford basic energy needs.

Furthermore, additional environmental and ethical concerns arise from the extraction of raw materials for batteries and renewable energy technologies, which can have adverse environmental impacts and ethical implications. Common environmental issues associated with mineral extraction for use in battery manufacturing include resource depletion and environmental degradation. On the other hand, such mining operations often can involve exploitation of labor, often affecting vulnerable communities globally.

Additionally, workers in fossil fuel-dependent industries face job displacement without guaranteed opportunities for reskilling or employment in the emerging clean energy sector. This is yet another reason why the economic shift from fossil fuels to an electrified economy requires careful management to prevent widespread unemployment and economic hardship in certain regions.

0.4 Proposed Approach

To address these challenges, a balanced and comprehensive strategy is necessary, focusing on both technological solutions and ethical considerations.

Firstly, the modernization of the electric grid is imperative. Investing in smart grid systems is essential to upgrade the grid by incorporating automation, advanced sensors, and multi-directional power flow capabilities. Smart grids enhance efficiency, reduce outages, and facilitate the integration of distributed energy resources (DERs) like rooftop solar panels and small-scale wind turbines. A phased implementation approach should be adopted to gradually introduce grid enhancements, allowing for adaptation and minimizing disruptions. Prioritizing investments in grid upgrades that enhance transmission capacity and resilience against cyberattacks and extreme weather events is crucial. Shifting from a centralized power generation model to a decentralized system will reduce energy loss during transmission and increase grid flexibility, empowering consumers to generate and manage their own electricity.

Secondly, the expansion of renewable energy infrastructure must be accelerated. Policy incentives such as tax credits, subsidies, and grants should be provided to encourage the development of renewable energy projects. Incentivizing both large-scale initiatives and

localized solutions, such as community solar projects, will facilitate broader adoption. Public-private partnerships can be established to collaborate with private companies in funding and deploying significant projects like battery production facilities that support the renewable energy supply chain. Streamlining regulatory processes by simplifying permitting and regulatory hurdles will accelerate the deployment of renewable energy installations.

Thirdly, the development of energy storage solutions is vital. Investment in research and development is necessary to advance cost-effective, high-capacity energy storage technologies to address the intermittency of renewable energy sources. Grid-scale storage deployment involves installing large-scale batteries and other storage systems to ensure a stable electricity supply during peak demand periods and enhance grid reliability. Developing standards for storage technology integration into the grid will facilitate widespread adoption.

Enhancing EV charging infrastructure is also essential. A nationwide expansion of EV charging stations should be undertaken, focusing on equitable distribution across urban, suburban, and rural areas. Implementing maintenance protocols and performance standards will ensure charging stations are functional and dependable. Encouraging innovative solutions, such as fast-charging stations and wireless charging, will improve convenience and reduce charging times.

Looking at electrification through an ethical lens, promoting equity in energy access is a key consideration. Targeted subsidies can offer financial assistance to low-income households for adopting energy-efficient appliances, EVs, and renewable energy systems. Inclusive planning should prioritize infrastructure projects in historically underserved communities to bridge the energy access gap. Involving local stakeholders in planning will ensure solutions meet the specific needs of different communities.

Ethical governance and policy reform are necessary to ensure transparent policies that prioritize public welfare over corporate interests, preventing monopolistic practices. Including local communities, businesses, and environmental groups in decision-making processes will ensure policies are equitable and address diverse concerns. Implementing oversight mechanisms will maintain accountability and fairness throughout the transition process.

Finally, investment in education and workforce training is crucial. Public awareness campaigns can educate the public about the benefits, challenges, and practicalities of

electrification and decarbonization to foster acceptance and participation. Incorporating sustainability and clean energy topics into educational programs will prepare future generations. Workforce development programs can offer training and reskilling opportunities for workers transitioning from fossil fuel industries to roles in the renewable energy sector.

0.5 High-Level Conclusion

The national goal of electrifying the economy is ambitious but achievable with a concerted effort that addresses both technological and ethical challenges. Modernizing the electric grid, expanding renewable energy infrastructure, and enhancing energy storage are critical technical steps. Simultaneously, promoting equity, ethical governance, and education ensures that the transition benefits all segments of society and mitigates potential negative impacts.

By adopting this comprehensive approach, the United States can move toward a sustainable, electrified future that enhances environmental stewardship, economic vitality, and social well-being. The successful realization of an electrified economy will position the nation as a leader in combating climate change and set a global example for sustainable development.

1. Introduction

Contributors of this report worked effectively in a multi-disciplinary team to understand the global and societal impacts of an electrified economy. Through collaborating with members from various engineering disciplines, they successfully chose a problem, conducted thorough research, and integrated findings into a cohesive report. Regular meetings, idea sharing, and draft refining were crucial in producing a well-organized final report that reflects true teamwork.

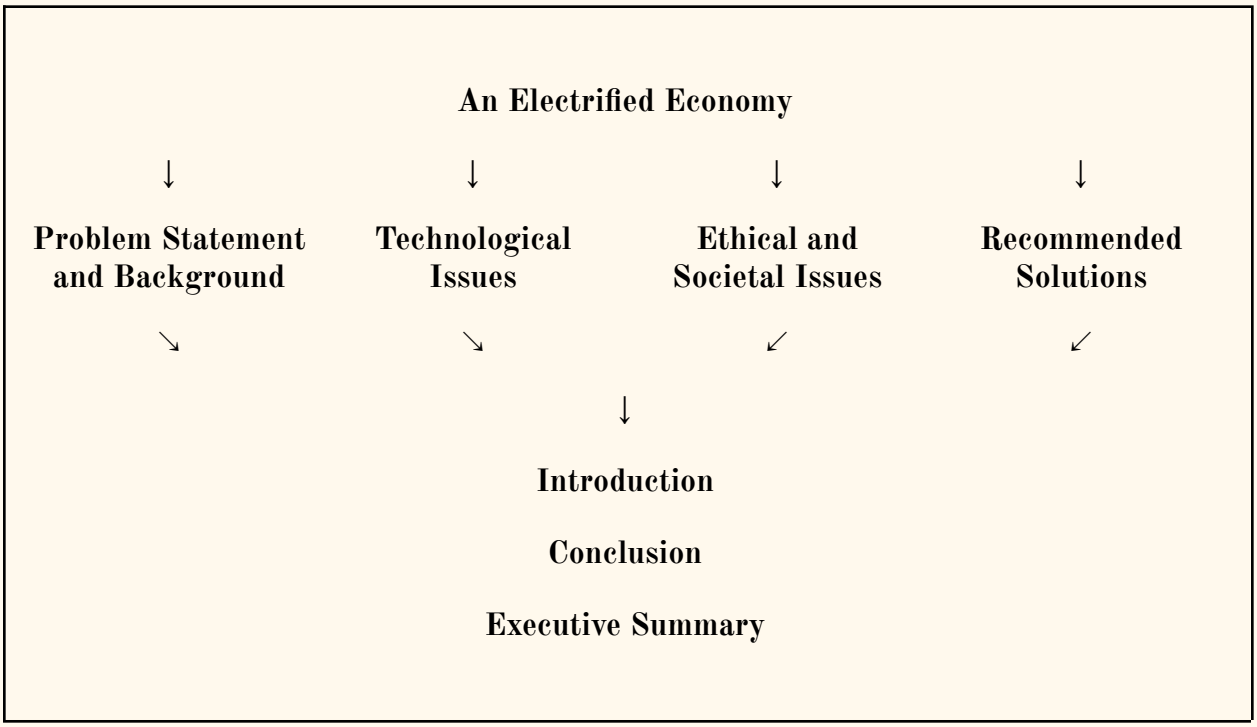
1.1 Methodology

The methodology for evaluating the feasibility of an electrified economy involved a multi-step approach. The approach started with conducting comprehensive literature reviews, analyzing articles, reports, and case studies, and gathering relevant data. Then, the project is broken down into fundamental elements and distributed among the team members to specialize in specific areas. Findings were discussed collaboratively to identify

key technological, societal, and ethical challenges. Based on the analysis, feasible solutions that address these challenges were developed, focusing on modernizing grid infrastructure, expanding renewable energy, improving battery technology, and ensuring equity in energy access and policy implementation.

1.2 Work Breakdown Structure (WBS)

The figure below is this report's Work Breakdown Structure (WBS). The project is broken down into fundamental elements as shown.



1.3 Responsibility Assignment Matrix (RAM) Chart

The figure below is this report's Responsibility Assignment Matrix (RAM) Chart, illustrating which team member was responsible for what part of the report.

	Pranav	Sahil	Troy	Yash
Front Matter	**D**	**P**	**R**	**D**
Executive Summary	**P**	**R**	**D**	**D**

Introduction	**R**	**D**	**P**	**D**
Problem Statement and Background	**D**	**D**	**R**	**P**
Technological Issues	**D**	**D**	**P**	**R**
Ethical and Societal Issues	**D**	**P**	**R**	**D**
Recommended Solutions	**D**	**R**	**D**	**P**
Conclusion	**P**	**D**	**D**	**R**
References Cited	**R**	**P**	**D**	**D**
Final Proofreading	**D**	**R**	**P**	**D**

P = Primary Responsibility for Writing
R = Review for content, form, and grammar
D = Discuss

2. Problem Statement and Background

2.1 Background on Electrifying the Economy

Before exploring the complexities of electrification and assessing whether it is feasible or even worthwhile to electrify the economy, it is essential to first define the term and understand what it means to *electrify* an entire economy.

An electrified economy is simply a society that adopts an economic system where electricity is its primary source of energy. In an electrified economy, sectors or industries that have traditionally relied on fossil fuels such as coal, oil, and gas are effectively replaced by electricity from **non-polluting, renewable** sources. (Rewiring America, n.d.). A common misunderstanding associated with such a transition is with our current source of electricity. Our society is more reliant on electricity than decades prior, which may suggest we're not far from the goal of complete reliance on electricity. However, the illusion present stems from our dependence on fossil fuels in power plants.

It is, therefore, important to clarify that in an electrified economy, fossil fuels are not just replaced with electricity, but with **clean (non-polluting), renewable** electricity (from sources like solar, wind, and hydro-electric power). In concert with it acting as a decarbonization strategy to reduce greenhouse gas emissions, combat climate change, and increase energy sustainability, electrification also offers the benefit of lowering the cost of electricity and democratizing access to energy (What is Electrification?, n.d.).

The road to an electrified economy is complex and multi-faceted, where numerous traditional systems and processes will be rendered obsolete. Transportation, for instance, will transition from gasoline and diesel-powered vehicles to electric vehicles, completely reshaping the fuel infrastructure (gas stations to charging stations, etc.) and supply chains. Industrial processes will adapt as well, with electrified tools and machinery quickly taking over factories. The combustion engine will be a thing of the past and replaced with electric alternatives, and fossil fuels will be reserved for during times of emergencies only. Residential and commercial buildings would embrace the power of electrification through renovations such as installing electric air and water heating and advanced energy management systems, in an effort to reduce the reliance on non-renewable energy sources. In agriculture, farming equipment and irrigation systems will pivot towards technologies capable of harnessing the electric power infrastructure established nationwide.

As is evident, the entire economy will go through a massive transformation and these transitions require significant upgrades to the power grid, advancements in energy storage technologies, and new strategies for managing increased electricity demand to ensure resilience and sustainability of the electrified economy.

2.2 National Goals for Electrification and Decarbonization

According to the Department of Energy, the national goals for electrification and decarbonization are 100% clean electricity by 2035, and "net-zero economy-wide greenhouse gas (GHG) emissions by no later than 2050" (DOE, 2023). These goals reflect a comprehensive strategy to combat climate change and global warming, improve energy security, and stimulate economic growth through clean energy innovation.

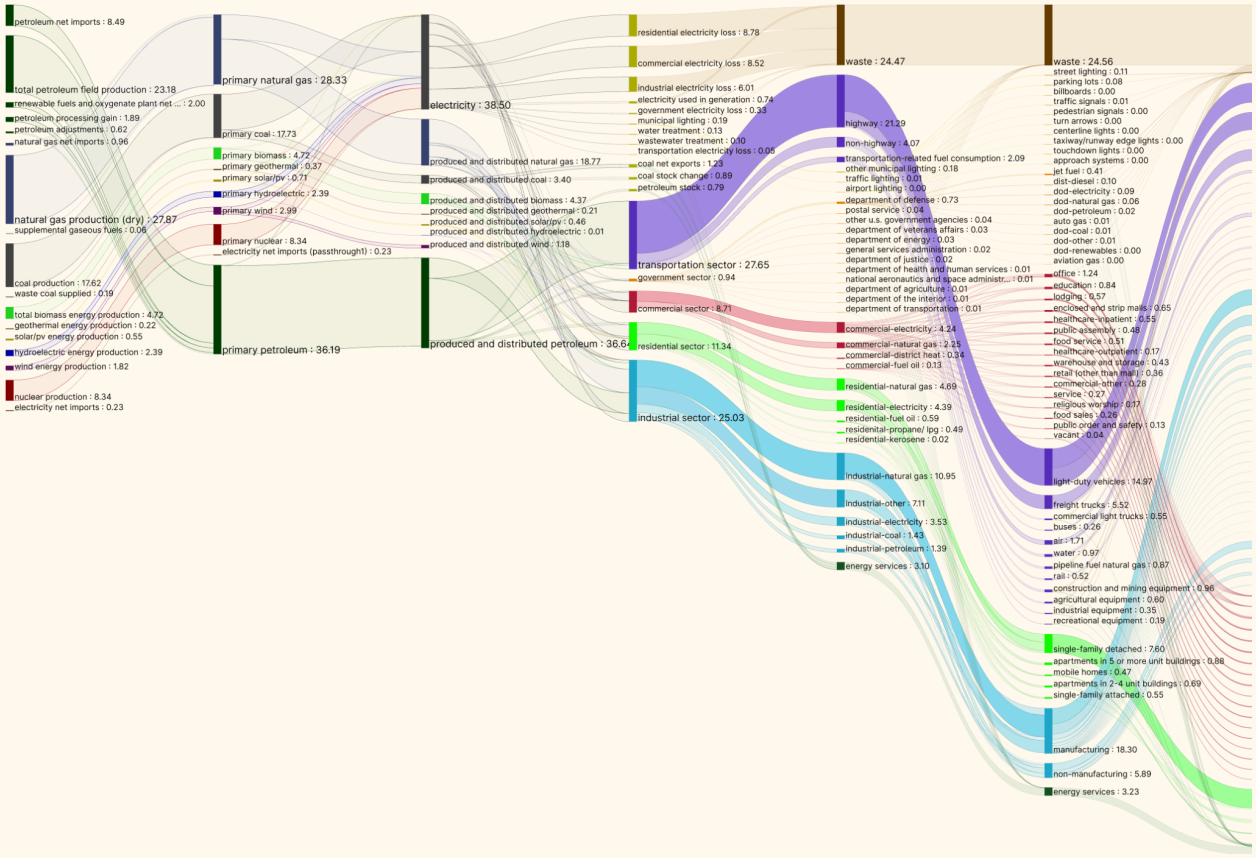


Figure 2.2.1: A snapshot of the Sankey diagram illustrating U.S. energy flows, compiled using integrated data from the EIA, RECS, MECS, CBECS, and NHTS. (Energy Literacy, 2024)

In order to answer the question "*Can the national goal of electrifying the economy (electric cars, reduction of fossil fuel use, etc.) be achieved?*", we must first look at the current state of energy usage in the U.S., from production to consumption. According to Figure 2.2.1, petroleum accounts for 36.19% of energy input, natural gas accounts for 28.33%, coal for another 17.73%, and nuclear for 8.34%. This leaves a little under 10% of all energy supplied through renewable energy sources such as solar, wind, and biomass.

Furthermore, we should also look at how this energy is used within the economy. The transportation sector accounts for 27.65% of total energy consumption, the commercial sector for 8.71%, residential sector for another 11.34%, and the industrial sector for 25.03%. It is also interesting to note that 24.47% of this energy ends up as waste.

This, therefore, provides a good starting point to help us answer how the national goal for electrification and decarbonization can be achieved.

2.3 Current Energy Infrastructure and Limitations

The existing energy infrastructure in the United States is a vast and intricate system that has evolved over more than a century. It encompasses power generation facilities, transmission and distribution networks, and the regulatory frameworks that govern them. While this infrastructure has reliably supported economic growth and societal needs, it was primarily designed around centralized, fossil fuel-based energy production. As the nation aims to transition toward an electrified and decarbonized economy, several limitations within the current infrastructure need to be addressed.

2.3.1 Dependence on Fossil Fuels

As portrayed in Figure 2.2.1, the United States derives most of its energy from fossil fuels such as petroleum, natural gas, and coal. As such, the current energy infrastructure is also primarily focused on producing electricity using these fuels. Such infrastructure is in the form of oil refineries, pipelines, coal plants, etc. This heavy reliance on fossil fuels contributes to greenhouse gas emissions and stands in the way of reaching the national decarbonization goals mentioned above. Transitioning away from fossil fuels will require not only replacing energy generation capacity but also reconfiguring the supply chains and industries built around them.

2.3.2 Aging Grid Infrastructure

The current U.S. electric grid was built over half a century ago (in the 1960s-70s) and while periodic renovations have improved this grid with automation and emerging technologies, it is increasingly less apt at meeting modern electricity needs. About 70% of transmission lines are stated to be over 25 years old and approaching the end of their useful life. This certainly poses a deepening risk to communities with an increased frequency of “power outages, susceptibility to cyber-attacks, or community emergencies caused by faulty grid infrastructure” (Grid Deployment Office, 2023). Nevertheless, we can utilize this opportunity to replace these parts of the electric grid with alternatives better suited to support and sustain an electrified economy — one that does not rely on a centralized power generation model and is optimized for clean, renewable energy.

2.3.3 Centralized Power Generation Model

The existing transmission infrastructure as it currently stands is built around a centralized, fossil-fuel-based power generation model. As a result of this architecture, energy loss increases proportionally as distance from power-plants increases due to resistive heat in the transmission lines — a direct result of the Joule Effect. Due to this physical property of transmission lines, 8-15% of energy is lost in this manner (Schonek, 2013). The same property is responsible for the current energy infrastructure being severely limited in integrating with renewable energy sources such as solar, wind, and hydroelectric as they are often located far from population centers. This creates a significant area of energy wastage and lowers the overall efficiency of renewable sources.

The centralized nature of the power generation model significantly contributes to grid inflexibility. This model relies on a one-way flow of power from large power plants to end users, making the system vulnerable to disruptions. When these centralized plants go offline, widespread outages occur. A more decentralized approach through Distributed Energy Resources (DERs) can enhance grid resilience. DERs, such as rooftop solar panels, small-scale wind turbines, and energy storage systems, allow for localized power generation and distribution, reducing the impact of outages and improving overall grid reliability (Distributed Energy Resources for Resilience, n.d.).

2.3.4 Vulnerability to Extreme Weather

The legacy energy infrastructure is increasingly vulnerable to extreme weather events, such as earthquakes, wildfires, and hurricanes. This is because such events can cause parts of the infrastructure such as power-plants to go offline due to increased stress on the infrastructure and thus creates a single point of failure. As a result, widespread power outages are becoming more common and the current grid's resilience to such events is inadequate to withstand such threats (Jacobo, 2024).

2.3.5 Lack of Energy Source Autonomy

As a direct consequence of the current energy infrastructure, consumer autonomy in choosing their preferred source of energy to power their homes, businesses, and municipalities is significantly limited. Most energy users are tied to the fossil-fuel-based energy mix provided by their local utility companies. This lack of choice restricts them

from independently transitioning to cleaner, electric alternatives. Thus, many users cannot individually aid in the national goal of electrification and decarbonization, and reduce the burden on the government, even if they are willing to invest in such alternatives.

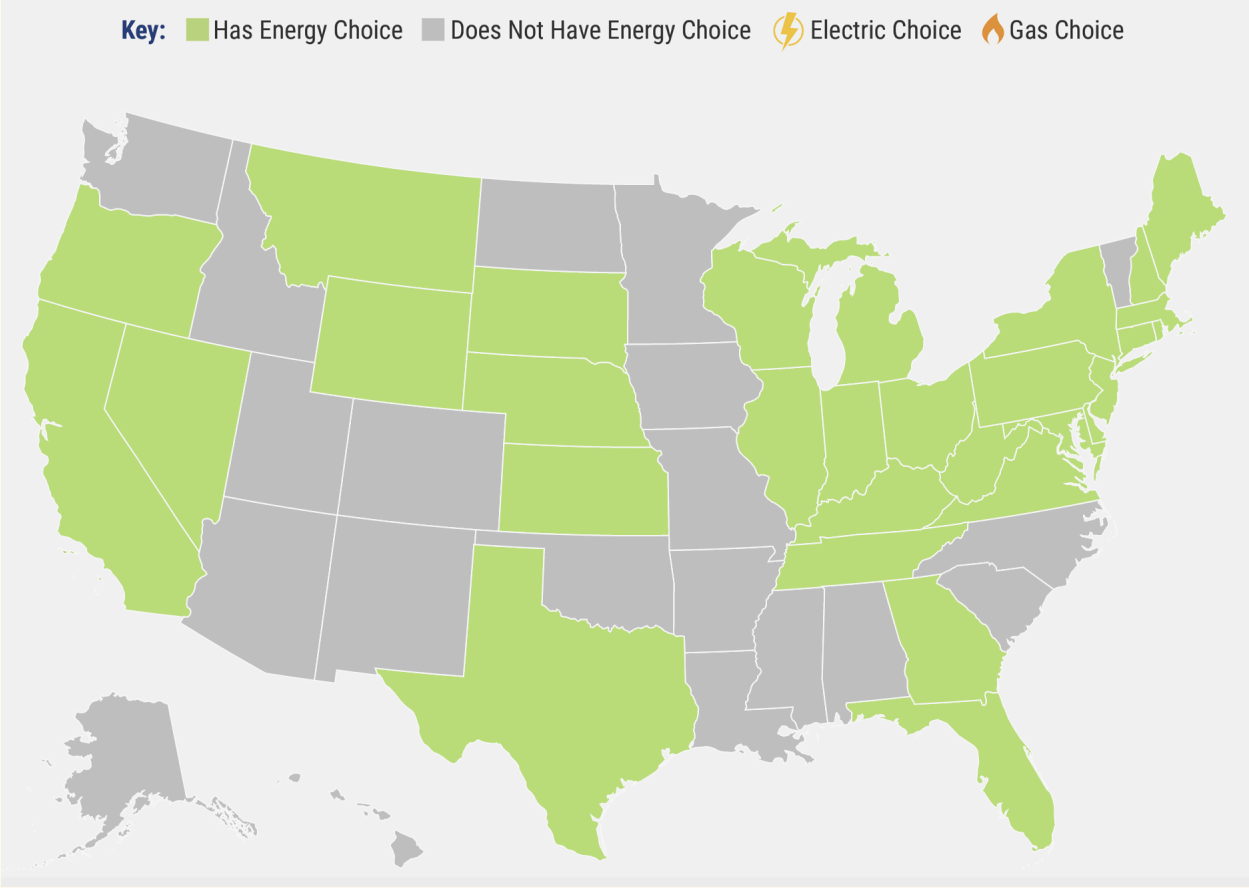


Figure 2.3.5.1: A geographical view of which states allow energy choice in any form. (ACCES, 2024)

2.3.6 Insufficient Energy Storage

Energy storage is an essential part of the energy infrastructure and helps balance supply and demand of energy. This is especially useful for renewable energy sources due to their inherently intermittent nature (wind energy is only available when the wind is blowing, solar is only available when the sun is shining, etc.). The lack of large-scale, efficient energy storage systems that are currently in place is, therefore, a significant limitation in our energy infrastructure.

Without adequate storage, excess renewable energy often goes unused and subsequently wasted. This needs to be addressed in order to maintain the supply of energy during times of high demand and reduce the probability of outage.

3. Technological Issues

Achieving an electrified economy presents significant technological challenges that must be addressed to ensure a successful all-electric economy and reduction of fossil fuels. Key issues include lack of renewable energy captured, aged power grid systems, and the insufficient electric vehicle (EV) infrastructure. These technological hurdles demand innovative solutions, strategic investments, and modernized infrastructure to support a reliable and sustainable electrified economy. In addition, the cost and scalability of these implementations are also important matters that must be considered. This section will cover the current situations, challenges, and potential solutions of renewable energy, grid infrastructure, battery technology, electric vehicle, material supply chain and mining, and costs and scalability.

3.1 Renewable Energy

In 2023, renewable energy accounted for about 30% of global electricity generation, marking an increase from 20% in 2021. Wind and solar energy were the largest contributors to this growth, driven by falling costs and favorable policies. Solar capacity grew by 48% in 2023 alone, largely propelled by expansions in China and the United States (IEA 2023). Despite this progress, fossil fuels still dominate overall energy consumption, particularly in sectors like transportation and heating. Many countries are setting ambitious targets to increase renewable energy capacity, aiming for net-zero emissions by 2050. However, while renewables are expanding rapidly in electricity generation, other sectors lag behind in transitioning away from fossil fuels.

Despite growth in renewable energy, significant challenges persist. Geographic variability affects the viability of wind and solar energy, with regions having differing potentials for renewable generation. Intermittent energy production—due to reliance on weather conditions—necessitates robust storage systems to maintain grid reliability. Moreover, integrating renewables into existing power grids designed for centralized fossil fuel plants creates grid instability risks during peak usage (World Economic Forum 2023). Developing nations face additional barriers, including a lack of infrastructure and

financing constraints. Furthermore, extreme weather events, like heatwaves and cold snaps, pose challenges to renewable output and grid resilience, slowing widespread adoption.

Improving energy storage is critical to overcoming renewable energy's intermittency. Advanced storage technologies like pumped hydro storage, compressed air energy storage, and lithium-ion batteries can help stabilize energy supply. Innovations in grid infrastructure, including smart grids with real-time monitoring, can optimize renewable integration (IEA 2023). Research into more efficient and cost-effective solar panels, wind turbines, and next-generation energy technologies can also accelerate adoption. Investments in energy diversification, combining solar, wind, and emerging technologies like green hydrogen, can mitigate geographic limitations. Global cooperation and funding for developing countries are essential to achieving equitable renewable energy expansion.

3.2 Grid Infrastructure

The U.S. electric grid infrastructure is facing significant challenges due to its aging design, much of which dates back to the 1960s and 1970s. Approximately 70% of transmission lines are over 25 years old and nearing the end of their 50-80-year lifespan. As electricity consumption reached a record 4.05 trillion kWh in 2022, the grid is struggling to meet the increasing demand driven by electrification and renewable energy integration (DOE 2023). The transition to clean energy has exacerbated these issues, with delays and high costs for renewable projects seeking to connect to the grid. Meanwhile, climate-induced extreme weather events, such as wildfires and hurricanes, have highlighted the grid's vulnerability, resulting in costly outages and maintenance challenges.

The primary challenges for grid infrastructure include ensuring stability under growing peak loads and enhancing resilience to extreme weather. Geographic and demographic changes have made power delivery less efficient, while delays in connecting new renewable energy projects pose significant hurdles. For example, wind and solar projects often face multi-year backlogs and prohibitive interconnection fees, making them economically unfeasible (Edge 2023). Additionally, climate threats such as wildfires, hurricanes, and flooding continue to damage critical infrastructure, causing widespread blackouts and economic disruptions. The U.S. grid must be expanded by 60% by 2030 to accommodate electrification goals, a costly and ambitious target given the regulatory and logistical complexities.

Modernizing the grid requires investment in smart grid technologies, real-time monitoring, and infrastructure upgrades. Smart grids can enhance reliability by identifying faults early and redistributing power efficiently. Grid modernization initiatives, supported by programs like the Grid Resilience and Innovation Partnerships (GRIP), aim to strengthen infrastructure against climate threats and reduce costs for consumers (DOE 2023). Solutions like microgrids and distributed energy resources can localize power supply, improving resilience during outages. Additionally, regulatory reforms to streamline renewable energy interconnection and investments in energy storage technologies, such as battery systems, are critical for balancing intermittent supply.

3.3 Battery Technology

Lithium-ion batteries currently dominate the battery market, powering everything from smartphones to electric vehicles (EVs). In 2023, global demand for lithium-ion batteries exceeded 700 GWh, a significant increase from previous years, driven by the surge in EV adoption and renewable energy storage needs (IEA 2023). China remains the leading producer and processor of lithium-ion batteries, controlling around 75% of global battery manufacturing capacity and two-thirds of lithium refining operations. Major companies such as CATL and BYD play pivotal roles in supplying the world with advanced battery technologies. Meanwhile, other nations, including the U.S. and Europe, are racing to develop their own battery production capacities, with numerous gigafactory projects underway to reduce dependence on Chinese supply chains.

A primary challenge in battery technology is improving energy density to extend usage range while reducing charging times. Despite technological advancements, lithium-ion batteries still face limitations in performance and safety, particularly at high capacities and during rapid charging. Safety concerns, such as thermal runaway leading to fires or explosions, remain critical, especially for large-scale storage systems (IEA 2023). Additionally, the environmental and geopolitical issues tied to lithium mining and refining are significant. China's dominance in refining and supply chains introduces vulnerabilities for other nations, creating risks of market disruptions. Furthermore, sourcing key materials like cobalt, often extracted under ethically questionable conditions, poses social and environmental challenges.

Emerging technologies like solid-state batteries offer promising alternatives, with higher energy density, faster charging times, and improved safety due to the absence of flammable liquid electrolytes (IEA 2023). Companies and researchers are also exploring

sodium-ion batteries, which utilize more abundant and cheaper materials than lithium-ion batteries, potentially reducing production costs and supply chain risks. Advances in battery recycling technology could alleviate pressure on raw material extraction and reduce environmental impact. Furthermore, efforts to diversify battery manufacturing by expanding gigafactory projects in the U.S. and Europe aim to reduce dependence on China and enhance energy security.

3.4 Electric Vehicle

Electric vehicle (EV) adoption is accelerating globally, with 14 million new electric cars registered in 2023. This growth is driven by supportive policies, expanding manufacturing, and consumer incentives. China leads the global market, with over 7 million EVs sold in 2023, representing 60% of worldwide sales. The United States and Europe also saw robust growth, with 1.4 million and 3.2 million new EV registrations, respectively (IEA 2024). Public charging infrastructure is expanding, with 2.7 million public charging points worldwide in 2022, mainly concentrated in China, Europe, and the US. However, the distribution of chargers remains uneven, with urban centers more equipped than rural areas.

Despite growth, the EV sector faces significant challenges. The lack of sufficient charging infrastructure, particularly in rural and underserved regions, limits adoption (IEA 2024). Additionally, the need for faster charging times and greater battery range hampers widespread use. High costs associated with battery manufacturing and vehicle purchase remain a barrier, particularly in developing economies. Extreme weather, such as cold winters, can also reduce battery performance and charging efficiency, affecting reliability. Finally, reliance on limited resources like lithium for batteries raises sustainability concerns.

Expanding charging infrastructure, especially in rural areas, is critical. Governments and private sectors can collaborate to deploy more charging stations and improve maintenance services. Innovations like solid-state batteries offer higher energy density, faster charging, and enhanced safety, while sodium-ion batteries provide a cost-effective alternative using more abundant materials (IEA 2024). Smart grid integration can optimize EV charging during non-peak hours. Subsidies and incentives for EV purchases and battery research can further support growth. Investing in renewable energy sources to power EVs will enhance the environmental benefits of electrification.

3.5 Material Supply Chain and Mining

The global transition towards sustainable energy is fundamentally reliant on the mining industry for sourcing the necessary raw materials. While crucial, the industry faces considerable challenges in aligning with sustainability goals. This essay will explore the current situation of the mining industry, its inherent challenges, and potential solutions for achieving sustainable mining practices that support the global energy transition.

The mining industry is instrumental in supplying the raw materials essential for building the components of renewable energy technologies like solar panels and wind turbines, as well as the batteries that power electric vehicles. This demand is reflected in the escalating mining activities globally, driven by the rising need for specific minerals and metals. The increasing adoption of electric vehicles (EVs), with nearly 14 million new EV registrations worldwide in 2023, exemplifies the growing need for materials, especially those used in battery production (IEA 2024). Similarly, expanding renewable energy infrastructure requires a consistent and substantial supply of diverse materials, all sourced through mining activities.

However, the mining industry encounters significant hurdles in harmonizing its practices with the sustainability objectives of the energy transition. The extraction, processing, and transportation of raw materials require a substantial amount of energy, and this energy is often derived from fossil fuels (Kirk 2022). This dependence generates considerable greenhouse gas emissions, contradicting the primary aim of the energy transition: to reduce emissions. The reliance on fossil fuels undermines the sustainability of mining operations. Moreover, mineral resources are finite, prompting concerns about the long-term viability of continuous extraction to satisfy the escalating demand for materials. To mitigate these challenges, a multifaceted approach is required to ensure a sustainable supply of materials for the energy transition.

Encouraging the recycling and reuse of materials, especially critical minerals found in batteries, can substantially reduce the demand for newly mined resources. Shifting towards a circular economy model, where materials are recovered and reused, is vital for a more environmentally responsible approach (Milman 2024). Adopting responsible mining practices is paramount. This includes using technologies and methods that minimize waste, reduce water consumption during extraction and processing, and prioritize responsible land management. Reclaiming mined lands to restore ecosystems is essential for mitigating long-term ecological impact. Ensuring ethical material sourcing is critical.

The industry needs to prioritize fair labor practices, safe working conditions, and minimize the exploitation of vulnerable communities. Initiatives that trace the origin of minerals and ensure adherence to ethical and sustainable standards are essential for promoting transparency and accountability.

By implementing these solutions, the mining industry can become a more sustainable sector that effectively supports the global energy transition. A shift towards responsible and sustainable mining practices is crucial not only for reducing environmental impact but also for promoting social responsibility and equitable development in the communities where mining takes place. As the world progresses toward a cleaner and more sustainable energy future, ensuring a responsible and sustainable source of materials will be essential to the success of this global transition.

3.6 Costs and Scalability

The pursuit of an electrified economy necessitates confronting critical reliability and performance challenges that threaten the stability and dependability of our future energy systems. Existing power grids, often characterized by aging infrastructure and centralized models, are increasingly vulnerable to disruptions from intensifying extreme weather events and the growing risk of cyberattacks.

The vulnerability of current energy systems to extreme weather is undeniable. In 2023, a winter storm in New England impacted 730,000 individuals, and in 2017, Hurricane Irma left 3,500,000 people in Florida without power (IEA 2024). These events underscore the urgent need to fortify energy systems, particularly with climate change projections indicating an escalation of such extreme weather events. The current U.S. electric grid, with 70% of its transmission lines over 25 years old, faces significant reliability issues. Extreme weather events can damage power lines, disrupt fuel supplies, and lead to cascading outages across critical infrastructure and services.

As power grids become more interconnected, the threat of cyberattacks grows, potentially causing blackouts, disrupting energy markets, and compromising national security. The interconnected nature of modern power grids creates vulnerabilities that sophisticated cyberattacks can exploit, potentially disrupting power distribution on a massive scale.

There are many potential strategies to address these challenges and build a more resilient grid. Transitioning from a centralized power generation model to a decentralized system, incorporating Distributed Energy Resources (DERs) like rooftop solar panels and

microgrids, can enhance resilience (Milman 2023). DERs allow for localized power generation and distribution, reducing reliance on centralized power plants vulnerable to disruptions. Implementing time-of-use pricing encourages consumers to shift energy consumption to off-peak hours, reducing strain on the grid during peak demand periods. Modernizing the electric grid by investing in smart grid technologies is crucial. Smart grids use sensors, automation, and data analytics to improve grid efficiency, anticipate and respond to demand fluctuations, and enhance system reliability.

The successful transition to an electrified economy depends on addressing these reliability and performance challenges. Through collaboration between governments, utilities, and consumers, along with investments in grid modernization and a focus on cybersecurity, a dependable, adaptable, and sustainable electrified future is possible.

4. Ethical and Societal Issues

4.1 Differences in Transitioning Renewable Electrification in Rural America

Historically, rural areas have lagged behind advances in electricity. Before the New Deal establishment of the Rural Electrification Administration, rural communities had a staggering gap in their access to electricity compared to urban areas: Only 10% of farmhouses had access to electricity compared to 90% access by non-farm rural and urban houses (Sablik, 2020). As a result of the REA's policies, today, power co-operatives service 42 million people, primarily in communities with no more than 10,000 residents, representing 60% of the US's landmass (Yañez-Barnuevo, 2023). Besides being massively successful in its objective, the New Deal policies have established systems that are fortunate to be resilient to the modern-day effort of transitioning to renewable energy technologies. A mix of grid failures due to severe weather and the push to be financially independent has pushed for distributed generation, meaning more co-op-owned power sources and a move away from reliance on a centralized source. One New Mexico-based company has expressed enthusiasm over this recent trend, elaborating how in-house energy systems could now be more tailored to account for the specifics of their climate and environment (Wagman, 2019).

Co-ops offer the benefits to consumers of greater alignment in their business practices compared to larger utility companies which border along monopoly status. Pacific Gas and Electric Company, or PG&E, has gained a track record of disregard of their consumers resulting from their business strategy to fund the construction of transmission lines rather

than make needed infrastructure upgrades to infrastructure that is 20 years older than the industry average (Environmental Working Group, 2022). Several wildfires were a direct result of this conscious neglect, including the most fatal California wildfire ever–the Bush Wildfire of 2018 (Romo, 2020)–and the largest single-event wildfire by area–the Dixie Wildfire of 2021, which destroyed nearly 1 million acres of land (Wiley, 2024)–among many other fires in the last 10 years which they are responsible and liable for (NPR, 2022). Likely to maintain its market share, PG&E pushed back against state-funded incentives for homeowners to install solar panels, demonstrating how profitability can hinder progress toward the goal of electrifying the economy. The trend towards distributed power generation, especially in the form of smaller cooperative companies, not only allows for greater democratic input for utility upgrades and modifications by consumers but also reduces the risk of a single power plant failure straining the power grid for a large group of people.

Co-operative electric companies, being owned by their customers and not-for-profit, would then seem to be less likely to misconduct and therefore be a better practical solution for rural Americans. However, considering that co-ops service 92% of all persistently poor counties (St. John, 2024), which are predominantly rural (National Association of State Energy Officials, 2020), the transition toward renewable energy systems is balanced by two opposing forces: enthusiasm and financial viability. One of the largest factors towards rural adoption of renewable energy technologies is income level (De Socio, 2023), which may result in rural development of renewable energy architecture being stifled by economic factors. A 2024 article identified that co-ops remain more reliant on coal for power generation than the rest of the country, and the cost to transition would be higher for these companies than urban utility companies (St. John, 2024). Combining the financial difficulty of rural areas with the gap in achieving 100% renewable energy in rural areas paints a pessimistic picture of the achievability of meeting the nationwide goal of removing dependence on fossil fuels.

Optimistically, recent incentives in the Inflation Reduction Act have subsidized $7.3 billion to the goal of adopting renewable energy technologies, doing so by "[accelerating] the closing [...] of coal-fired generation [facilities]," allowing upgrades to existing plants in the form of carbon-capture technologies, and funding renewable energy development projects (St. John, 2024). This initiative comes at an opportune time, as renewable energy has become economically favorable long term compared to conventional coal-based power plants, coupled with political factors limiting the amount of fossil fuel supply to the US. Policy analyst Michelle Solomon has urged to capitalize on the unique opportunity

presented, one which allows for the U.S. to meet its elusive goal of nullifying its carbon emissions while being economically and politically favorable (Milman, 2023).

4.2 Inequities in Urban Electrification

4.2.1 Hindrances in Electric Vehicle (EV) adoption

As represented by the current shift in the U.S. electric vehicle (EV) market targeting the luxury car brand niche, the EV market has found footing in a market niche targeting high-income customers. A large majority of individuals who own EVs are well-educated, have high incomes, and are homeowners (Hardman et al., 2021). The EV market struggles to break into a larger clientele, with only 7.9% of EV owners in 2017 belonging to the middle-income classification (Hyun Lee et al., 2019). This implicates a slow replacement of gas and diesel-powered vehicles with electric-powered vehicles, thus slowing the overall goal of electrifying the economy. One important, self-reinforcing factor that contributes to this is public EV charging infrastructure. From a business strategy point of view, creating charging stations in low-income areas generates fewer potential customers than building the station in an area where residents have more available income. The lack of public charging stations reduces the certainty that lower income individuals will buy an EV, as they tend to have no dedicated parking and may not have options to park at work. Even if chagrin stations are near enough to benefit lower-income areas, they have poor reliability, such as found by UC Berkeley's finding that 25% of San Francisco Bay Area charging stations did not function properly (Morgan, 2023). Considering that EV charging infrastructure is a large determining factor for lower-income individuals' purchase of EVs, the current market favors building charging stations in wealthier areas or to avoid their installation altogether, which bottlenecks the widespread adoption of EVs.

While government incentives exist to remedy this, they have limited benefit to their target beneficiaries. Incentives for buying EVs are only available in certain states and are limited to new EVs from certain dealerships. These incentives thus are limited to subsidizing market-price EVs and do not offer benefits for their resale. As a result, most of the benefits are reaped by high-income consumers, whose motivations to buy EVs are unchanged by these incentives. In contrast, low-income consumers have a larger share of consumers who purchase cars from the resale market and not through dealerships (Hardman et al., 2021). In order to spur the adoption of EVs, incentives must be better suited to benefit more sectors of the population. Encouraging middle to lower income consumers through legislatively-funded incentives would span a much larger audience, not

only enabling their ability to afford EVs, but also encouraging industry to create infrastructure and products to meet their now increased consumer base. As most current EV owners treat this purchase as an alternative, rather than a replacement, to their gas-powered cars (Morgan, 2023), motivation to create this infrastructure is low also in considering the other factors previously mentioned. Government incentives must be delicately, however, to avoid favoritism among companies and vehicle types.

Ultimately, the widespread adoption of EVs begins with increasing the viability of city cars for a common populace. Doing so would spark the self-sufficiency of the EV sector by creating demand for used EVs and different chassis types. Doing so would revitalize the desire for electrifying other modes of transportation–such as buses, trains, and eventually aircraft–to meet the norm of transportation refueling.

4.2.2 Difficulties in Electrifying Urban Buildings

Electric appliances have the potential to benefit leasers and homeowners in reducing the cost to operate their appliances over the long-term. However, a challenge of electrification is in the transition towards these technologies. Santa Monica city, which has a population just under 90,000 residents (US Census, n.d.), estimated that the cost of electrifying the entire city to be half to all of $1 billion (Casuso, 2023). Evidently, this figure poses a challenge towards transitioning the city, let alone the country. Santa Monica boasts a 58% larger median income than the national average and around the same poverty rate (US Census, n.d.; Shrider et al., 2021), suggesting that the city represents a wealthier populace than the national average. The burden of cost faced by Santa Monica will be less than that of poorer, more populous, and older cities.

Sentiment and cost factors may still influence individual building owners to upgrade their infrastructure to fully-electric appliances. However, the cost to purchase the appliances, especially when required more immediately, to replace an old appliance, is off-putting. In order to offset the cost of installation, property owners may increase rent prices to tenants. This scenario becomes more probable in considering the appeal of being an early adopter of these appliances as a way of attracting higher income tenants. As a result, the number of safe, affordable options available to lower income tenants may become limited, thus offsetting the burden to them down-the-line.

4.3 Societal Health-Related Concerns of Electrification

4.3.1 Health Benefits Associated with Reducing Emissions

A positive aspect associated with the goal of electrification are the health benefits associated with the reduction of emissions. The goal of nullifying our carbon emissions has been modeled by a 2019 study to improve the overall health of urban areas in California (). Deaths associated with low air quality can be prevented through the overall reduction of emissions. Moreso, the decreased health toll would translate to reduced economic burden in treating air-quality related health issues. Low-income urban residents, who tend to live closer to areas of low air quality, would benefit from reduced negative health impacts and the costs associated with treating them. Reasonably, these groups are more enthusiastic about reducing emissions than their counterparts (Hardman et al., 2021). However, as mentioned in section 4.2.1, financial factors limit their ability to contribute toward the electrification transition, thus suffer the adverse health effects of continued reliance on fossil fuels in urban environments. A push to implement business-friendly solutions to allow for lower-income transition into the electrified economy would thus also see financial benefits through the reduction of financial burden on the medicine industry, thus reducing the amount of time required to break even on this investment.

4.3.2 Health Dangers Associated with Employment in Renewable Energy Technologies

Maintenance of renewable energy infrastructure would become a larger workforce requirement in an electrified economy. However, currently, these jobs carry considerable amount of risk that would require innovations in practices, standards, and technology. As the number of wind turbines increase, the total number of turbine technician accidents have grown consistently from 1998 to 2017. Wind turbine technicians risk equipment malfunction which could result in death from falling large heights, fires caused by arcing, or strangulation due to harness misuse or malfunction (Evans, 2018; OSHA, 2011). Solar panel maintenance also carries the risk of electrical arcing or discharge (Evans, 2018). The risk posed by increasing the number of power farms in the form of turbines or solar panels is the fact that many of these farms would be far away from city centers, thus increasing the response time of emergency services. In tandem with the goal of adopting more renewable energy technologies, more research into updated safety protocols or emergency response procedures should be conducted to reduce the deaths and injuries associated with accidents among maintenance professions.

4.4 Supply Chain Tolls

In order to develop the infrastructure necessary for renewable energy sources, developers must extract resources and modify landscapes to meet the needs of the projects. As mentioned in section 3.6, the cost of acquiring materials is considerable. Mining to extract the materials necessary for solar panels can lead to water contamination and pollution due to the machinery required to extract the materials (Hoffs, 2022). As many raw materials are sourced from developing countries, which can sometimes prioritize the output of these mining operations over the safety of its communities and laborers. One Madagascar mining operation was even attributed to introducing a poisonous invasive species near to their operations, damaging the ecology and poisoning many nearby denizens (Business & Human Rights Resource Centre, n.d.).

The end-of-life practices for these technologies also bear concern, given that many turbine blades and solar panel hazardous waste will end up in landfills after their use. The waste estimated to be produced by the solar industry after their use varies, but one group of researchers estimated that the waste produced would be larger than the amount produced by our personal electronics (Shellenberger, 2021).

5. Recommended Solution

As is portrayed clearly in section 2.2, there is a long way to go to achieve the national goals of electrification and decarbonization. This requires a comprehensive approach that addresses the technological limitations while also considering the ethical and societal implications of this undertaking. Thus, we present a series of recommended solutions (technical and ethical) that can help the United States keep up with its ambitious goals.

5.1 Technical Recommendations

5.1.1 Modernization of the Electric Grid

Smart grid systems are capable of managing increased demand and integration with renewable energy sources. It incorporates automations, advanced sensors, and multi-way distribution of electricity to facilitate decentralized power generation and distribution. This not only improves efficiency, but also reduces outages. Therefore, modernizing the electric grid with smart grid systems is a great step to achieving energy autonomy and resilience.

Since these systems introduce an extensive change to the current electric grid, these systems need to be slowly phased in to allow for gradual adaptation of infrastructure, including prioritized investment in grid upgrades. These upgrades can be in the form of smart meters, grid-scale batteries, DERs, and advanced monitoring systems. This will boost the transmission capacity of the grid, and improve resilience of the grid by offering power distribution flexibility. We should also layer in security and strengthen protections against cyberattacks through robust encryption, firewalls, and real-time threat detection systems (DOE, 2024).

The integration of DERs, such as rooftop solar panels, will provide individuals with agency by granting them greater control over their electricity production and consumption. As a direct result, the average cost of electricity will decrease and individuals will have greater access to the energy marketplace — further democratizing energy.

5.1.2 Investment in Renewable Energy Infrastructure

Transitioning away from fossil fuels and to renewable energy will reduce greenhouse gas emissions and promote sustainable development. Projects like the Gigafactory are pioneering in this space and can considerably aid in the national goals of electrification and decarbonization. In fact, according to Tesla's calculations, it will only take 100 Gigafactories to "transition the whole world to sustainable energy" (gohd, 2017).

We need to accelerate the deployment of renewable energy projects in wind, solar, and hydroelectric power. This can be achieved methodically and equitably by providing policy incentives such as tax credits and subsidies for energy developers and adopters alike. Collaboration between the government and the private sector can be leveraged to fund and deploy large-scale projects such as the Gigafactory for the welfare of the general public. Likewise, localized solutions can also be promoted through government incentives for community solar projects and small-scale wind farms in rural areas.

5.1.3 Development of Energy Storage Solutions

As mentioned above, energy storage is a critical bottleneck with renewable energy and thus, if we are to move to a carbon-neutral future and fulfill the national goals in this space, great investment in grid-scale energy storage solutions and technologies is necessary. The development and deployment of such solutions will address the

intermittency of renewables and will ensure stable electricity supply during periods of peak demand (IEA, 2023).

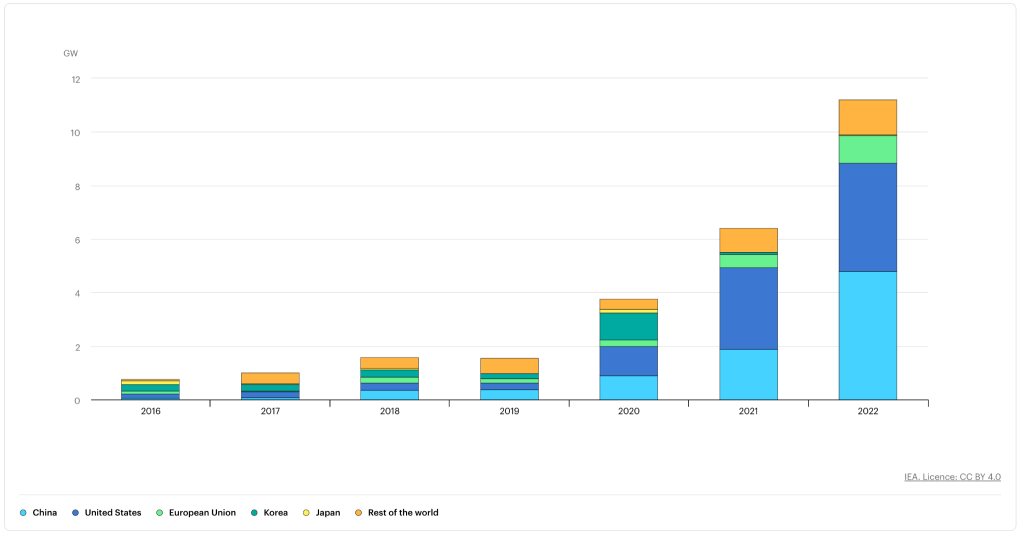


Figure 5.1.3.1: Annual grid-scale battery storage additions (in GW), 2017-2022 (IEA, 2023)

As is evident in Figure 5.1.3.1, the annual grid-scale battery additions are increasing throughout the world, however, to accommodate renewable energy to completely take over in the new electric grid, there needs to be a concerted effort and investment to fund research and develop cost-effective, high-capacity energy storage systems. This will, again, require government assistance through grants, subsidies, etc.

A key form of energy storage solutions that is of particular interest and will significantly aid in reaching the goal of electrification and decarbonization is the integration of charging stations. This is of strategic importance as transportation accounts for 27.65% of all energy usage in the US. Therefore, investing and deploying charging stations throughout the country will bridge the gap between gas and electric vehicles and incentivize the adoption of EVs. There is great progress being made in this endeavor with about 192,000 charging stations publicly available currently in the United States with another 1,000 being added every week, according to reports (FHWA, 2024).

5.2 Ethical Recommendations

5.2.1 Promoting Equity in Energy Access

Historically, energy access has been riddled with inequalities and has not considered the ramifications on low-income communities that have been left behind with about 1 in 7 families living in energy poverty in the US (Mills, 2023).

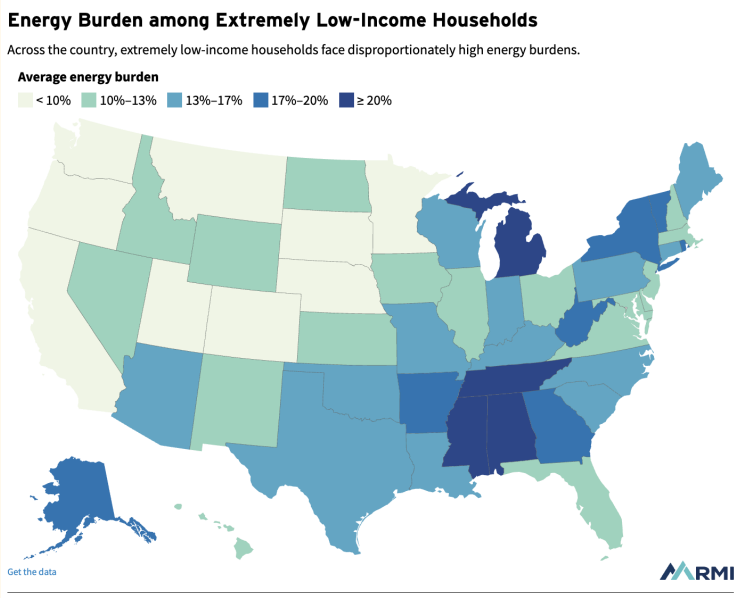


Figure 5.2.1.1: Energy Burden among Extremely Low-Income Households (Mills, 2023)

Considering that moving towards an electrified economy will be a massive disruption, with great upside and downside for several stakeholders such as new and legacy energy companies, the end-user, and the government, it is important to ensure that the benefits of electrification are equitably distributed among all socioeconomic groups, especially underserved communities.

While this is an opportunity to right the wrongs of energy poverty, without a deliberate effort towards an equitable outcome, this transition into the electrified economy could in-fact exacerbate the existing inequalities and continue to hurt low-income households.

We, therefore, recommend targeted subsidies and assistance programs that offer financial support for underserved communities and that reduce the friction in adopting energy-efficient appliances and renewable energy systems. Furthermore, the government can practice inclusive infrastructure planning that prioritizes renewable energy projects and grid upgrades in historically marginalized and underserved areas.

5.2.2 Ethical Governance and Policy Reform

As aforementioned, the electrified economy brings with it great disruption in various sectors including but not limited to transportation, industrial, and commercial sectors. It is, therefore, important to establish transparent and inclusive policies for energy system upgrades, and decarbonization initiatives, else risk the potential for negative socio-economic outcomes.

As such, we recommend the participation of local communities, businesses, and environmental groups in the planning and decision-making processes. It is also important that the policies prioritize public welfare over corporate interests and for there to exist strict oversight mechanisms to prevent monopolistic behavior by corporations. This ethical governance structure will ensure accountability, fairness, and stakeholder participation to minimize potential negative externalities.

5.2.3 Invest in Education

It is crucial for future generations to understand the importance of, and continue the work towards an electrified economy. By developing and funding education programs that increase the public's awareness and understanding about electrification and decarbonization, we can empower individuals and communities to participate in electrifying the economy.

Investing in public awareness campaigns and integrating topics related to clean energy, sustainability, and electrifications into academic curriculum will lead to mass acceptance, address the misconceptions about renewable energy, and encourage informed decision making.

Since the blue-collar workers in fossil-fuel-dependent industries will be disproportionately impacted by this transition, it is important to invest in workforce training programs. This, in addition to community workshops and seminars, can offer specialized training for careers in renewable energy, grid modernization, and energy storage technologies. Moreover, support for research and development by funding academic and private research via grants can further advance clean energy technologies and their practical applications.

6. Conclusion

Achieving a fully electrified economy in the United States involves reimagining decades-old infrastructure and fundamentally altering the production, distribution, and consumption of energy. The shift away from fossil fuels toward renewable electricity is not merely a technological challenge; it is a comprehensive societal undertaking. This includes modernizing an electric grid originally designed for large, centralized power plants into one capable of supporting distributed energy resources and accommodating the inherently intermittent nature of solar, wind, and other renewables. As illustrated, integrating robust energy storage systems—from battery arrays to emerging technologies in physical storage—is essential to ensure a stable, continuous supply of electricity, even as weather patterns fluctuate and seasonal demand peaks.

The complexity of electrification in the particular context of transportation also demands careful attention to the reliability and ubiquity of electric vehicle charging infrastructure. Without consistent access to well-maintained, strategically located charging stations, broad EV adoption will remain constrained—particularly in rural or low-income areas that often lag behind in infrastructure investments. Overcoming this shortfall requires not only public-private partnerships and market-based incentives but also policy frameworks that prioritize equity, ensuring that the benefits of electrification do not concentrate solely in affluent enclaves.

Simultaneously, ethical and societal considerations must guide the course of action. The restructuring of energy systems must not exacerbate social inequalities. For instance, as the essay highlighted, historically underserved communities risk bearing a disproportionate share of the financial and infrastructural burdens if targeted support is not provided. Workforce training for individuals transitioning out of fossil fuel industries is equally crucial to prevent economic displacement and to ensure that those whose livelihoods once depended on legacy energy sectors find new opportunities in the clean energy economy. Transparent governance, community-level input, and the careful allocation of subsidies and incentives can facilitate a just transition. This holistic approach promises not only cleaner air and healthier communities but also a more resilient national grid, better prepared to withstand natural disasters and cyber threats.

The likely near-term future will see incremental advances, with some states and municipalities leading by example through pilot projects, early policy innovations, and community-driven renewable energy initiatives. Over time, as grid upgrades proliferate, energy storage costs fall, and charging networks expand, these successes can be replicated and scaled, hastening the pace of national progress. While achieving complete

electrification and decarbonization by mid-century remains a formidable challenge, these gradual improvements—informed by interdisciplinary research, stakeholder engagement, and balanced policy—offer a promising path forward. It may be neither swift nor uniform, but with persistent effort, adaptive strategies, and a commitment to fairness and sustainability, the United States can transform its energy landscape into one defined by cleanliness, resilience, and inclusivity.

7. References

2024 Smart Grid System Report (2024). U.S. Department of Energy https://www.energy.gov/sites/default/files/2024-02/2024%20Smart%20Grid%20System%20Report_untagged.pdf

Business & Human Rights Resource Centre (n.d.). Human Rights in the Mineral Supply Chains of Solar Panels. Business & Human Rights Resource Centre. https://www.business-humanrights.org/en/from-us/briefings/transition-minerals-sector-case-studies/human-rights-in-the-mineral-supply-chains-of-solar-panels/

Casuso, J. (2023, March 6). Transitioning to Electricity Could Cost Property Owners Up to $1 Billion. Santa Monica Lookout. https://www.surfsantamonica.com/ssm_site/the_lookout/news/News-2023/March-2023/03_06_2023_Transitioning_to_Electricity_Could_Cost_Property_Owners_Up_to_1_%20Billion.html

De Socio, M. (2023, November 22). Why Rural Americans Are (and Aren't) Adopting Renewable Energy. CNET. https://www.cnet.com/home/energy-and-utilities/why-rural-americans-are-and-arent-adopting-renewable-energy/

Distributed Energy Resources for Resilience. (n.d.). Energy.gov. https://www.energy.gov/femp/distributed-energy-resources-resilience

Energy Literacy. (2024). Departmentof.energy. http://www.departmentof.energy/

Environmental Working Group (2022, October 13). Safety investigation faults PG&E's failure to maintain aging infrastructure against wildfire risks. Environmental Working Group. https://www.ewg.org/news-insights/news-release/2022/10/safety-investigation-faults-pges-failure-maintain-aging

Evans, S. (2018, September 6). What are the most dangerous jobs in the energy sector? Power Technology. https://www.power-technology.com/features/most-dangerous-jobs-in-the-energy-sector/?cf-view

gohd, C. (2017, April 16). Elon Musk: 100 Tesla Gigafactories Could Power the Entire World. Futurism. https://futurism.com/elon-musk-100-tesla-gigafactories-could-power-entire-world

Grid Deployment Office. (2023, October 19). What does it take to modernize the U.S. electric grid? Energy.gov. https://www.energy.gov/gdo/articles/what-does-it-take-modernize-us-electric-grid

Hardman, S. et al. (2021, August 30). A perspective on equity in the transition to electric vehicles. MIT Science Policy Review. DOI: 10.38105/spr.e10rdoaoup. https://sciencepolicyreview.org/2021/08/equity-transition-electric-vehicles/

Hoffs, C. (2022, October 19). Mining Raw Materials for Solar Panels: Problems and Solutions. Union of Concerned Scientists. https://blog.ucsusa.org/charlie-hoffs/mining-raw-materials-for-solar-panels-problems-and-solutions/

Hyun Lee, J. et al. (2019, September). Who is buying electric vehicles in California? Characterising early adopter heterogeneity and forecasting market diffusion. Energy Research & Social Science, Vol. 55, pp. 218-226. https://doi.org/10.1016/j.erss.2019.05.011. https://www.sciencedirect.com/science/article/pii/S2214629618312258?via=ihub

IEA (2023), Annual grid-scale battery storage additions, 2017-2022, IEA, Paris https://www.iea.org/data-and-statistics/charts/annual-grid-scale-battery-storage-additions-2017-2022, Licence: CC BY 4.0

IEA. (2023, July 11). Energy storage. IEA. https://www.iea.org/energy-system/electricity/grid-scale-storage

INVESTING IN AMERICA: Number of Publicly Available Electric Vehicle Chargers Has Doubled Since Start of Biden-Harris Administration | FHWA. (2024). Dot.gov. https://highways.dot.gov/newsroom/investing-america-number-publicly-available-electric-vehicle-chargers-has-doubled-start

Jacobo, J. (2024, August 31). How extreme weather impacts power grids and the customers who rely on them. ABC News. https://abcnews.go.com/US/extreme-weather-impacts-power-grids-customers-rely/story?id=113210374

Kirk, K. (2022, October 24). Energy loss is single biggest component of today's electricity system» Yale Climate Connections. Yale Climate Connections. https://yaleclimateconnections.org/2022/10/energy-loss-is-single-biggest-component-of-todays-electricity-system/

Mills, R. (2023, December 18). 1 in 7 Families Live in Energy Poverty. States Can Ease That Burden. RMI. https://rmi.org/1-in-7-families-live-in-energy-poverty-states-can-ease-that-burden/

Milman, O. (2023, January 30). US renewable energy farms outstrip 99% of coal plants economically – study. The Guardian. https://www.theguardian.com/us-news/2023/jan/30/us-coal-more-expensive-than-renewable-energy-study

Morgan, K. (2023, November 8). Three big reasons Americans haven't rapidly adopted EVs. British Broadcasting Company. https://www.bbc.com/worklife/article/20231108-three-big-reasons-americans-havent-rapidly-adopted-evs

National Association of State Energy Officials (2020, May). Rural Data Resources for State Energy Planning and Programs. https://www.naseo.org/data/sites/1/documents/publications/Final%20Rural%20Report%20May%2020201.pdf

National Transmission Analysis Maps Next Chapter of US Grid Evolution. (2024). Nrel.gov. https://www.nrel.gov/news/features/2024/national-transmission-planning-study.html

Occupational Health and Safety Administration (2011, August 22). Accident Report Detail. OSHA, Report No. 200534501, Report ID 0751910. https://www.osha.gov/ords/imis/accidentsearch.accident_detail?id=200534501

On The Path to 100% Clean Electricity On the Path to 100% Clean Electricity 2. (2023). https://www.energy.gov/sites/default/files/2023-05/DOE%20-%20100%25%20Clean%20Electricity%20-%20Final.pdf

Rewiring America. (n.d.). Rewiring America. https://www.rewiringamerica.org/

Romo, V. (2020, June 16). PG&E Pleads Guilty On 2018 California Camp Fire: 'Our Equipment Started That Fire'. National Public Radio (NPR). https://www.npr.org/2020/06/16/879008760/pg-e-pleads-guilty-on-2018-california-camp-fire-our-equipment-started-that-fire

Sablik, T. (2020, Q1). Electrifying Rural America. Federal Reserve Bank of Richmond. https://www.richmondfed.org/publications/research/econ_focus/2020/q1/economic_history

Schonek, J. (2013, March 25). How big are power line losses? Schneider Electric Blog. https://blog.se.com/sustainability/2013/03/25/how-big-are-power-line-losses/

Shellenberger, M. (2021, June 21). Dark Side To Solar? More Reports Tie Panel Production To Toxic Pollution. Forbes.

https://www.forbes.com/sites/michaelshellenberger/2021/06/21/why-everything-they-said-about-solar---including-that-its-clean-and-cheap---was-wrong/

Shrider, E. A. et al. (2021, September 14). Income and Poverty in the United States: 2020 US Census. https://www.census.gov/library/publications/2021/demo/p60-273.html

St. John, J. (2024, September 12). How $7.3B will help rural co-ops build clean power—and close coal plants. Canary Media. https://www.canarymedia.com/articles/clean-energy/how-7-3b-will-help-rural-co-ops-build-clean-power-and-close-coal-plants

State By State Energy Choice Map | Competitive Energy. (2024, June 24). Competitive Energy. https://competitiveenergy.org/consumer-tools/state-by-state-links/

The Associated Press at National Public Radio (2022, April 14). The nation's largest utility agrees to pay more than $55 million for two wildfires. National Public Radio (NPR). https://www.npr.org/2022/04/12/1092259419/california-wildfires-pacific-gas-electric-55-million

United States Census. (n.d.). QuickFacts Santa Monica city, California. US Census. Accessed December 11, 2024. https://www.census.gov/quickfacts/santamonicacitycalifornia

What is Electrification? (n.d.). Energy.gov. https://www.energy.gov/electricity-insights/what-electrification

Wiley, H. (2024, April 14). PG&E hit with $225-million lawsuit for 2021 Dixie fire damages. Los Angeles Times, CA. https://www.latimes.com/california/story/2024-04-14/pg-e-hit-with-225-million-lawsuit-over-allegations-of-negligence-that-helped-spark-dixie-fire

Yañez-Barnuevo, M. (2023, June 12). Microgrids and Energy Improvements in Rural Areas. Environmental and Energy Study Institute. https://www.eesi.org/articles/view/microgrids-and-energy-improvements-in-rural-areas